Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester III Examination April 2011
NT802 Application of Nanomaterials

Date 22nd April 2011 Time: 10:00 to 10:30 Max. Marks 20

Part A

(Do as directed)

Q 1. Oligonucleotides are ______________. (1)
(i) Multistranded DNA (ii) Single stranded DNA
(iii) Double stranded DNA

Q2. . Anti bodies are ______________ created by immune systems of higher organisms that that can recognize a virus and act to destroy it. (1)
(i) carbohydrates (ii) nucleic acids
(iii) Protein molecules (iv) Amino acids

Q3. The unusual properties of individual gold atom are attributable to ______________ ______________ that stabilized the $6S^2$ electron pairs. (1)
(i) photoluminiscence (ii) Sonic boom
(iii) Relativistic effect

Q4. One of the essential requirements of high oxidation activities of gold nanoparticle is to have size (R) (1)
(i) $R < 4$ nm (ii) $4 \text{ nm} < R < 10$ nm
(iii) $R > 10$ nm

Q5. Compared to small sized quantum dots, large sized quantum dots emit ______________ wavelength. (1)
(i) Longer (ii) Shorter
(iii) Intermediate

Q6. Confinement factor of the optical mode in single quantum well laser is ______________ than that in multiple quantum well laser. (1)
(i) Larger (ii) Smaller
(iii) Equal

Q7. In the dynamic mode of AFM, the cantilever is driven at ______________. (1)
(i) At lower frequency (ii) At resonance frequency
(iii) At higher frequency

Q8. In the heat mode of AFM the cantilever is coated with a material having ______________ ______________. (1)
(i) Different electrical conductivity (ii) Different resonance frequency
(iii) Different thermal expansion coefficient

Q9. Compared to carbon fibers, carbon nanotubes have ______________ aspect ratio and ______________ tip radius of curvature. (1)
(i) High (ii) Small
(iii) Equal

Q10. ______________ studies the transport of ions rather than electrons in nanoscale systems. (1)
(i) Nanomechanics (ii) Nanolithigraphy
(iii) Nanoelectronics (iv) Nanoionics

Q11. Nanoelectronics refers to the use of nanotechnology on ________________ components. (1)

(i) Electronics (ii) Electrical engineering
(iii) Electronic engineering (iv) Engineering

Q12. Nanoelectronics holds the promise of making ______________________________ more powerful than are possible with conventional semiconductor fabrication techniques. (1)

(i) 64-bit (ii) Central processing unit
(iii) Reduced instruction set computer (iv) Microprocessor

Q13. Plasmonics is… (1)

(i) A field of nanophotonics that holds the promise of molecular-size optical device technology
(ii) The science of fluorescence nanoparticles used in modern fireworks
(iii) A hypothetical science used in science fiction weaponry (plasma canon)
(iv) The technology used to design and build the laser guided photonic gyroscopes used in aviation

Q14. The role of mesoporous titania film in dye sensitized solar cell is ______________________________ ______________________________. (1)

(i) Electron capturing and transport (ii) Light absorption
(iii) Antireflective coating

Q15. Nanobots are nanoscale robots… (1)

(i) Do not exists yet (ii) Exists in experimental form in laboratories
(iii) Are already being used in nanomedicines (iv) Will be used in the unmanned mission to Mars

Q16. What is grapheme? (1)

(i) A new material made from carbon nanotube
(ii) A one atom thick sheet of carbon
(iii) Thin film made from fullerenes
(iv) A software tool for graphical representation of nanoparticles

Q17. Quantum wells are formed in semiconductors by sandwiching a material between two layers of another material having wider ____________________. (1)

(i) Electrical conductivity (ii) Lattice parameters
(iii) Band gap

Q18. MEMS stands for…. (1)

Q19. The band gap of a semiconductor increases as the size of particle decreases, if the size of particle is ______________________________. (1)

(i) Below Debye length (ii) Above Bohr radius
(iii) Below Bohr radius (iv) Below lattice parameter

Q20. Dye sensitized solar cells use nanostructured ________________ (1)

(i) CdS (ii) PbS
(iii) Fe_2O_3 (iv) TiO_2

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Part B

(Write each section on separate answer book)

Section I

Q1. (a) What is self-assembly? (3)

(b) Explain how the molecular electronics based gas sensors works? (3)

(c) Describe different possible roles of nanobots in the field of nanomedicines. (4)

Q2. (a) Why antibodies and oligonucleotides are most widely being used in molecular recognization applications? (4)

(b) Explain how the molecules can be crafted in to working circuits? (6)

Section II

Q3. (a) Explain how the relativistic effect stabilizes the $6s^2$ electron pair in gold atom. (4)

OR

(a) Describe the factors affecting efficiency of luminescence from quantum dots. (4)

(b) With the help of schematic energy band diagram explain different types of quantum well lasers. (6)

Q4. (a) Explain the working of AFM cantilever based nanosensors in static mode. (5)

(b) Explain the working of AFM cantilever based nanosensors in dynamic mode. (5)

OR

(b) Explain the working of AFM cantilever based nanosensors in heat mode. (5)

Q5. (a) Why gradual degradation with time of the emission performance is more in the case of single wall carbon nanotube that that for multiwall carbon nanotube? (2)

(b) Explain the working of dye sensitized titania photovoltaic cell. (4)

(c) What is photonic band gap crystal? Also explain how it can serve as a lossless waveguide to confine or direct the propagation of light along a specific channel. (4)

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